

Chapter 1:

Introduction

Agriculture is the backbone of many economies, providing livelihoods for a large portion of the global population. As the world's population continues to grow, the demand for improved agricultural efficiency to meet food production needs has become more critical. Traditional farming methods often rely on manual labor and intuition, leading to inefficiencies, particularly in the use of water, a valuable and increasingly scarce resource. With advancements in technology, there is now an opportunity to modernize agricultural practices through automation and the Internet of Things (IoT), significantly improving productivity and resource management.

The Smart Crop Monitoring System, which utilizes the ESP32, Real-Time Clock (RTC), and cloud integration, presents an innovative solution to address these challenges. The system uses a combination of sensors, including soil moisture sensors, digital water flow sensors, and solenoid valves, to monitor and manage irrigation effectively. The ESP32, with its built-in WiFi and Bluetooth capabilities, enables seamless cloud connectivity, allowing farmers to remotely access real-time data. By automating irrigation and integrating real-time data monitoring, the system ensures optimal water usage while minimizing the need for manual supervision.

Cloud integration allows farmers to monitor field conditions remotely, accessing vital information about soil moisture, water flow, and environmental factors from anywhere. The Real-Time Clock (RTC) enables precise irrigation scheduling, ensuring that crops are watered at the right time and with the correct amount of water. This technology not only conserves water but also improves crop yield by providing plants with the ideal moisture levels, promoting more efficient and sustainable farming practices.

Chapter 2:

Literature Survey

The different Crop Monitoring techniques by various researchers are described in this section. Teddy Surya Gunawan et al,[1] Smart agriculture leverages IoT and wireless networks to enhance food security and productivity. Real-time monitoring of soil moisture, humidity, and temperature optimizes irrigation and reduces water waste. A mobile application enables remote access to sensor data for better crop management. This study compares automated and manual irrigation on chili plants, showing that automation maintains optimal moisture levels. The automated system accelerates plant growth compared to manual watering. Data-driven insights improve decision-making, leading to higher crop yields. Smart agriculture increases efficiency by reducing human intervention and optimizing resource use. Ultimately, this system enhances sustainability and productivity in farming. Deepthi P et al,[2] The study presents a smart farming monitoring system using fuzzy logic control to enhance agricultural productivity. It integrates sensors for temperature, humidity, soil moisture, light intensity, and water levels to optimize irrigation. Experimental results show 96.5% accuracy and 95.3% efficiency in predicting and running the water pump. The system responds within 1 minute to environmental changes and has a 98.7% pump operation reliability. Fuzzy logic outperforms traditional methods by improving precision, reducing waste, and minimizing labor costs. It enables sustainable farming under changing environmental conditions. Future advancements in sensors and AI will further enhance smart farming. Optimized farming systems will drive agricultural efficiency and sustainability. Arief Sudarmaji et al,[3] Efficient water management in coastal horticulture requires continuous soil moisture monitoring. Farmers face challenges in frequently measuring land humidity manually. This study designs a wireless soil moisture monitoring system using IoT. Capacitive soil moisture sensors, integrated with an ESP32, enable remote monitoring via Wi-Fi. The ESP32, programmed with Arduino, reads sensor data and connects to the internet. A web-based interface and Google Spreadsheet, developed using Node-RED, display real-time soil moisture data. The system also controls irrigation sprayers automatically via a water pump. This approach improves irrigation efficiency and optimizes water use in horticulture. Radhika R. Boralkar et al,[4] Agriculture is a crucial sector in India, relying heavily on

manual labor. To improve efficiency, automation is essential. IoT-based smart agriculture uses sensors like temperature, humidity, and soil moisture to monitor crops.

ESP32-based automated irrigation optimizes water usage. Real-time data helps farmers make informed decisions, reducing labor and resource wastage. Smart farming techniques enhance productivity and crop health. These innovations ensure better yields and sustainability. Implementing such systems makes agriculture more efficient and reliable. G. Karuna et al,[5] The Internet of Things (IoT) enables automation in various fields, including agriculture. Traditional farming techniques often fail to ensure efficient crop yields. Smart agriculture enhances productivity using sensors and data analytics. The system monitors soil moisture, temperature, and nutrients in real time. A Kodular-based application helps farmers track crop conditions remotely. It also recommends suitable crops based on soil type and season. This approach reduces manual labor and increases efficiency. Ultimately, it improves crop yields and boosts agricultural profitability. Berko et al,[6] During the dry season, minimal rainfall affects plant life, while excess water in the rainy season may harm crops. A system is needed to store and manage water efficiently. Ghanaian farmers require real-time crop data to improve productivity. IoT-based solutions can enhance traditional irrigation systems by integrating sensors to monitor temperature, pH, soil moisture, and humidity. These sensors send data to smart devices via the internet, enabling efficient farm management. Adopting modern technology is crucial for Ghana to remain competitive globally. Many farmers still rely on manual labor, reducing efficiency. This project explores a smart irrigation and plant monitoring system using IoT for better water management. Nazwa Amelia Purnama et al,[7] The global food crisis has become one of the most pressing issues requiring serious attention, especially as the world's population continues to grow. This growth has led to an increasing demand for food, while agricultural resources such as fertile land, water, and labor become increasingly limited. These conditions present significant challenges for farmers, including droughts that damage crops, floods that devastate agricultural lands, and pest attacks that significantly reduce yields. Furthermore, global climate change exacerbates the situation, directly impacting the stability of food production. Unpredictable weather patterns threaten harvest success, increase the risk of crop failure, and decrease agricultural productivity. Therefore, innovative solutions, such as utilizing Internet of Things (IoT)-based technologies, are urgently needed to support the sustainability of the agricultural sector by enhancing efficiency, effectiveness, and global food security. This study aims to contribute to addressing

these challenges through a modern technological approach. Gilroy P. Pereira et al,[8] This project introduces a smart IoT-based drip irrigation system using ESP32 to automate the watering process for crops. The ESP32 is connected to soil moisture, temperature, humidity, and water flow sensors to monitor field conditions in real time. When the soil is dry and the temperature is suitable, the ESP32 automatically opens a solenoid valve to start irrigation. The water flow sensor regulates irrigation time, preventing overwatering. A humidity sensor alerts the user when humidity levels are too high or too low. Farmers can remotely monitor and control the system using the Blynk app, ensuring efficient water management. The system was successfully tested and proved effective in growing green onions, demonstrating its potential for smart and sustainable farming. Gonesh Chandra Saha et al,[9] The IoT-based smart agricultural system enhances crop monitoring by continuously tracking temperature and soil moisture levels. It uses wireless sensors, actuators, and cloud-based analytics to provide real-time data for better farm management. The system helps regulate temperature to 26.2°C while maintaining soil moisture at 45%, ensuring optimal crop growth. Farmers can use this data-driven approach to make informed decisions, leading to higher yields and sustainability. By leveraging IoT technology, agriculture becomes more efficient and resource-conscious. The system's ability to automate monitoring reduces manual labor and optimizes water usage. This advancement can revolutionize global farming and help ensure food security amid climate change. Continued research and innovation will further improve smart agriculture solutions. Dr Lakshmi Narayana et al,[10] Agriculture has been practiced for ages and is both a science and an art. With advancing technology, agriculture also needs modernization. IoT plays a vital role in smart farming by providing real-time data. Sensors monitor humidity, temperature, moisture, and rain levels. The ESP32-based system automates irrigation when moisture drops. Data is wirelessly transmitted for remote monitoring. Temperature adjustments can be made based on crop type. This system improves efficiency, reducing manual labor and optimizing water usage. Anees Abu Snehneh et al,[11] This paper presents a hydroponics monitoring system that collects key parameters such as temperature, water level, pH, and nutrient levels. The system uses an ESP32 microcontroller along with sensors like TDS, pH, and temperature sensors. The ESP32 processes data and automatically controls water and nutrient pumps. Users can monitor parameters and control the system remotely via the Blynk smartphone app. The system activates pumps automatically if values deviate from the required range. The hydroponics monitoring system

was successfully designed and implemented. Experimental data was collected and analyzed. The results demonstrate the system's effectiveness in maintaining optimal growing conditions. Neema S et al,[12] This paper says IoT-based smart agriculture is a revolutionary approach that enhances farming efficiency using technology. Traditional farming relied on manual labor and outdated techniques, which were time-consuming and inefficient. This modern system integrates sensors and microcontrollers to monitor soil moisture, temperature, humidity, and motion. Optimizing water usage helps reduce resource wastage while meeting the growing food demand. The proposed system includes an ESP32, soil moisture sensor, humidity sensor, PIR motion sensor, flame sensor, and temperature sensor. It provides real-time data for better environmental monitoring and improved crop yield. Additionally, it can incorporate solar energy for sustainability.

Chapter 3:

Problem Analysis and Solution

3.1 Problem Statement

In modern agriculture, efficient crop management is essential to maximize yield while minimizing resource use. Traditional farming methods often lack real-time monitoring capabilities, leading to inefficiencies such as over-irrigation, inadequate nutrient management, and suboptimal growing conditions. Weather fluctuations, including unexpected rainfall, temperature extremes, and humidity variations, further complicate these challenges, impacting crop health and productivity. To address these issues, there is a need for a smart crop monitoring system that leverages technology to provide farmers with timely data on soil moisture, temperature, humidity, light conditions, and weather patterns. By utilizing an ESP32-based solution equipped with various sensors and cloud connectivity, this project aims to empower farmers with actionable insights. The ESP32's built-in Wi-Fi and Bluetooth capabilities enable seamless data transmission, allowing real-time monitoring and control. The implementation of such a system will enable proactive decision-making based on live environmental conditions, enhance crop health, and contribute to sustainable agricultural practices. Ultimately, this will support food security and environmental sustainability, ensuring that farmers can adapt to changing weather conditions effectively.

3.2 Proposed Solution

The Smart Crop Monitoring System addresses the challenges of inefficient water usage and labor-intensive farming by automating irrigation processes. Traditional farming often leads to over-irrigation or under-irrigation, wasting water and reducing crop yields. To solve this, the system uses ESP32 as the control unit, gathering data from soil moisture sensors, digital water flow sensors, and other components. The ESP32's built-in Wi-Fi capability enables seamless cloud integration, allowing farmers to monitor and control irrigation remotely. A Real-Time Clock (RTC) ensures accurate irrigation scheduling, activating solenoid valves when the soil moisture level falls below a set threshold.

The system allows farmers to set custom irrigation schedules via a keypad and provides realtime feedback on soil conditions through an LCD display. With cloud integration, farmers can access critical data anytime, anywhere using their smartphones or computers. By uploading sensor data to the cloud, they can track soil moisture, water flow, and environmental conditions in real time, making informed irrigation decisions.

This approach ensures that water is used efficiently, reducing wastage and ensuring crops receive the optimal amount of moisture. Automation reduces the need for manual labor, saving time and effort for farmers. The system enhances crop yields by optimizing irrigation, ensuring that crops are watered at the right time with the right amount of water. Overall, the Smart Crop Monitoring System improves water conservation, reduces labor costs, and increases agricultural productivity, offering a sustainable and efficient solution for modern farming.

Chapter 4:

Methodology & Implementation

4.1 Block Diagram

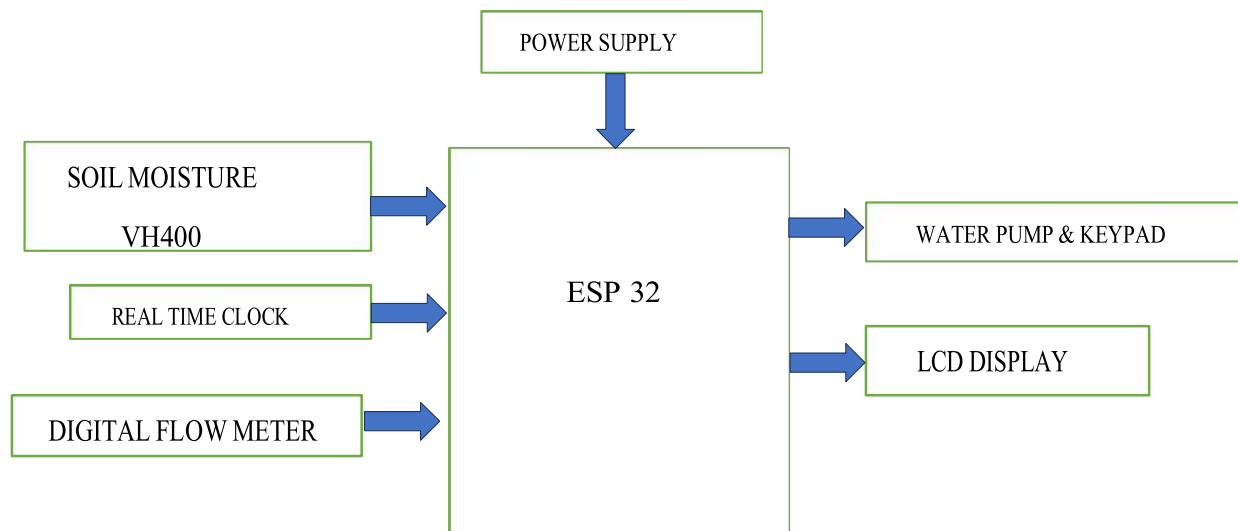


Fig (4.1) : Block Diagram of Smart Crop Monitoring System

4.1.1 Pictorial Representation



Fig (4.1.1) – Pictorial Representation

4.1.2 Structure of the Work

1. Sensing Soil Moisture:

- The Soil Moisture Sensor (VH400) continuously monitors the moisture level in the soil.
- It sends analog signals to the Arduino microcontroller, indicating the current moisture content.

2. Real-Time Clock Monitoring:

- The Real-Time Clock (RTC) provides accurate time and date information to the Arduino.
- This allows the system to schedule irrigation events based on specific times or intervals.

3. Digital Flow Meter Reading:

- The Digital Flow Meter measures the amount of water being delivered during irrigation.
- It sends digital signals to the Arduino, indicating the flow rate and total water volume.

4. ESP32 Microcontroller Control:

- The ESP32 acts as the brain of the system.
- It receives data from the soil moisture sensor, RTC, and flow meter.
- It processes this information to determine when and how much water is needed for irrigation.
- Based on the analysis, it controls the water pump and keypad to start/stop irrigation and adjust water flow accordingly.

- With built-in Wi-Fi, the ESP32 enables real-time data transmission to the cloud for remote monitoring and control.

5. User Interface and Display:

- The LCD Display provides a user interface for interaction and monitoring.
- It shows real-time information like soil moisture levels, current time, water flow rate, and total water usage.
- The Keypad allows users to manually control the system, set schedules, and adjust parameters.

6. Power Supply:

- The Power Supply provides the necessary power to all components of the system.

Overall Workflow:

1. The soil moisture sensor measures the moisture level.
2. The ESP32 reads the sensor data and compares it with the desired moisture threshold.
3. If the moisture level is below the threshold, the ESP32 activates the water pump to start irrigation.
4. The flow meter measures the water flow, and the ESP32 monitors the total water volume.
5. The RTC keeps track of time and schedules irrigation events based on user preferences.
6. The LCD display shows real-time data and allows user interaction.
7. The ESP32's built-in Wi-Fi enables cloud integration for remote monitoring and control via a smartphone or computer.

This system optimizes irrigation by ensuring that plants receive the right amount of water at the right time, saving water and energy while promoting healthy plant growth.

4.1.2 Flow Chart



Fig (4.1.2) – flow chart showing how the model works

Chapter 5: Result & Discussion

Result

This work presents the design and implementation of an automatic irrigation system utilizing an ESP32 microcontroller. The system incorporates a soil moisture sensor (VH400), a realtime clock (RTC), a digital flow meter, a water pump, a keypad, and an LCD display. The system successfully demonstrates the following functionalities:

- **Real-time Soil Moisture Monitoring:** The system accurately measures and displays soil moisture levels, providing valuable insights into the plant's water needs.
- **Scheduled Irrigation:** The RTC enables precise scheduling of irrigation events, ensuring that plants receive water at optimal times.
- **Automatic Water Delivery:** The system automatically triggers the water pump when soil moisture falls below a predefined threshold, ensuring adequate water supply to the plants.
- **Flow Rate Monitoring:** The digital flow meter accurately measures the water flow rate during irrigation, allowing for precise control of water usage.
- **Cloud Integration & Remote Monitoring:** The ESP32's built-in Wi-Fi enables realtime data transmission to a cloud platform, allowing farmers to monitor and control irrigation remotely.
- **User-friendly Interface:** The LCD display provides a user-friendly interface for monitoring system status, adjusting parameters, and viewing historical data. The keypad enables manual control and configuration.

By leveraging ESP32's wireless capabilities, this system enhances efficiency, reduces manual labor, and promotes sustainable water management in agriculture.

Discussion

- **Water Conservation:** By precisely controlling water delivery based on real-time soil moisture data, the system minimizes water wastage, leading to significant water conservation.
- **Improved Plant Health:** Consistent and timely water supply promotes healthy plant growth and development by preventing both under-watering and over-watering.
- **Reduced Labor Costs:** Automation eliminates the need for manual monitoring and irrigation, reducing labor costs and freeing up time for other tasks.
- **Enhanced Efficiency:** The system operates efficiently by delivering water only when and where it is needed, optimizing resource utilization.
- **Cloud Integration & Remote Access:** With ESP32's built-in Wi-Fi, farmers can monitor and control the irrigation system remotely, improving accessibility and convenience.
- **Data-driven Decision Making:** The system provides valuable data on soil moisture levels, water usage, and irrigation schedules, enabling informed decision-making and system optimization.

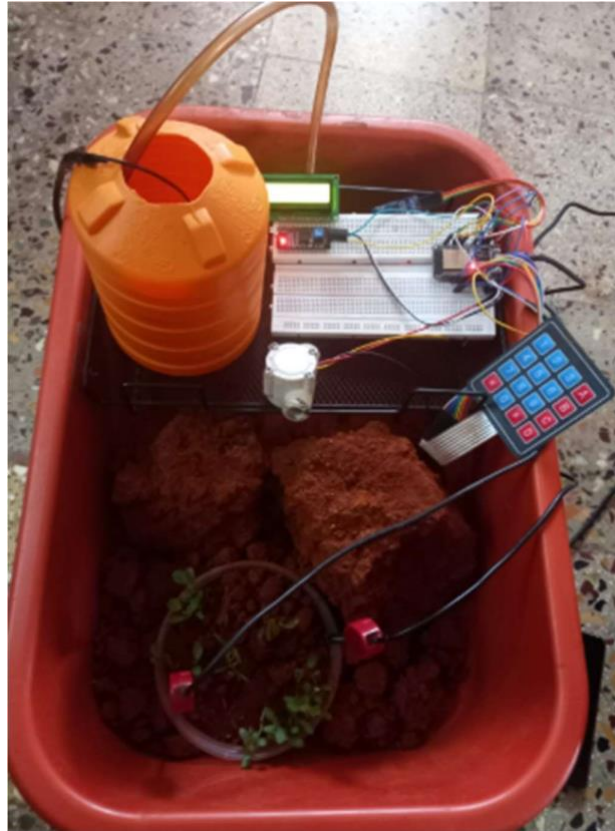


Fig. 5.1 – Implemented model picture

Chapter 6:

6.1 Conclusion

This project successfully demonstrates a functional automatic irrigation system. The system utilizes an ESP32 microcontroller to integrate components such as a soil moisture sensor, realtime clock, digital flow meter, water pump, keypad, and LCD display. By monitoring soil moisture levels in real-time, the system schedules and executes irrigation events, controlling water flow to optimize plant health and minimize water wastage. The built-in Wi-Fi capability of the ESP32 enables seamless cloud integration, allowing for remote monitoring and control. The user-friendly interface ensures easy system management, while real-time data access enhances decision-making. While the current implementation provides a strong foundation, future enhancements, including advanced weather integration, AI-driven

irrigation adjustments, and support for multiple irrigation zones, can further expand the system's capabilities and applicability in diverse agricultural and horticultural settings. This project showcases the potential of microcontroller-based solutions to address challenges in water management and promote sustainable agricultural practices.

6.2 Future Trends

- **Automated Irrigation:** Integrate a relay and water pump for fully automated irrigation based on real-time soil moisture data.
- **Solar Power:** Utilize solar panels to power the system, making it ideal for remote and offgrid agricultural areas.
- **AI & Predictive Analytics:** Implement AI-driven models to predict crop water requirements and optimize irrigation schedules based on historical and real-time data.
- **Mobile App:** Develop a mobile application for real-time monitoring, manual control, and system alerts, ensuring farmers can access data from anywhere.
- **Advanced Sensors:** Expand sensor capabilities to include pH, nitrogen, phosphorus, and potassium (NPK) sensors for precise soil health monitoring.

- Drone Integration: Utilize drones equipped with multispectral and thermal cameras for aerial crop monitoring, disease detection, and large-scale field analysis.
- Data Analytics Dashboard: Design a web-based or mobile dashboard for visualizing sensor data trends, historical irrigation records, and system performance metrics.
- Multi-Crop Support: Calibrate the system for different crop types, ensuring optimal irrigation and nutrient management for various plant species.
- Weather Integration: Integrate weather forecast APIs to adjust irrigation schedules dynamically based on upcoming rainfall, temperature, and humidity changes.
- Edge Computing: Enable local data processing on the ESP32 to reduce cloud reliance, improving response times and system reliability in areas with limited internet connectivity.
- Wireless Sensor Networks: Expand coverage by deploying multiple ESP32-based wireless sensor nodes to monitor soil conditions across large agricultural fields.
- Pest Detection: Implement image recognition technology using AI-powered cameras to detect pests and plant diseases, sending real-time alerts to farmers.

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APPENDIX A

Plagiarism Report:

PLAGIARISM SCAN REPORT

Report Generation Date: 16-02-25

Words: 994

Characters: 7151

Excluded URL : N/A

2% Plagiarism	98% Unique
1 Plagiarized Sentences	59 Unique Sentences

APPENDIX B

Research publication:

[https://ijsrem.com/download/ Smart Crop Monitoring System](https://ijsrem.com/download/SmartCropMonitoringSystem)

For DOI verification, Click here <https://www.doi.org/10.55041/IJSREM41569>

<https://ijsrem.com/volume09issue02february2025/>

